

ORF309/MAT380

**Brownian Motion I: Introduction
and Continuous Time Limit of RWs**

(pp. 135-138)

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Agenda for today

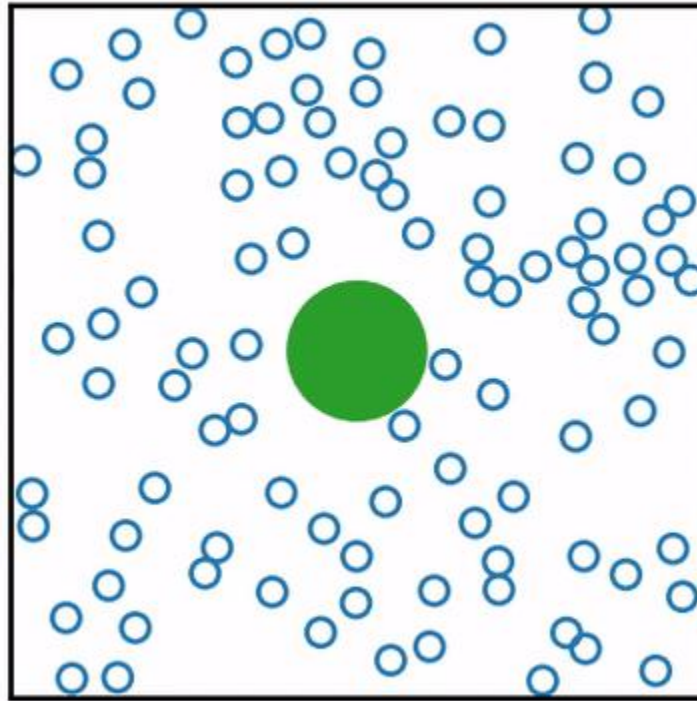
- Historical background
 - Brown's explanation
 - Einstein's explanation
- Mathematical model
 - Central Limit Theorem
- Example from stock market

Brief History of BM

- First observed by a Scottish **botanist** called **Robert Brown** – **1828**
 - Observations of pollen movement in a liquid
- Other scientists got interested in *mysteriously random motion...*
- But **Albert Einstein** studied it from a **Physics angle** and **explained mathematically** – **1905**
 - **All scientific research is interconnected and important!**
 - It helps us understand the world we live in
 - **Tyche** reveals some instantiations of the world



Brownian Motion in 2d

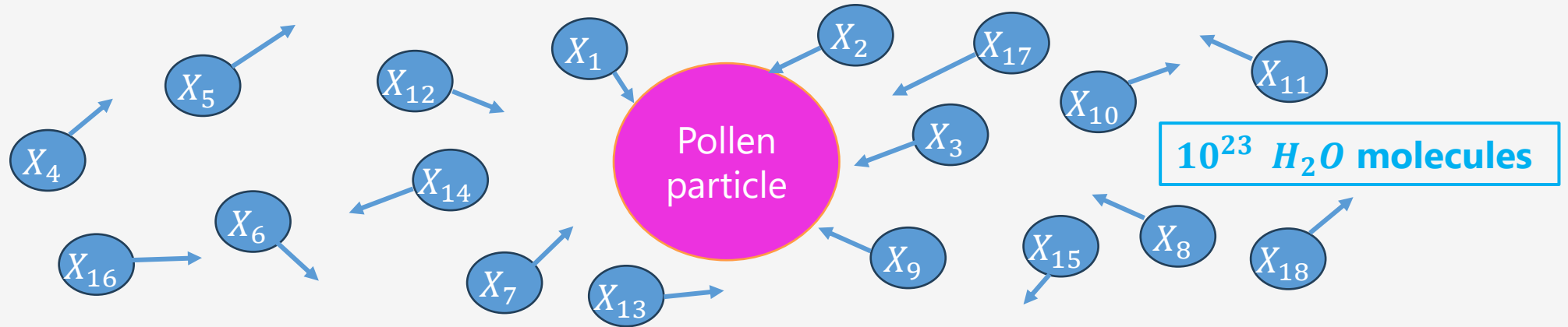


<https://scipython.com/blog/a-simple-two-dimensional-brownian-motion-animation/>

Einstein's explanation of BM - 1905



Elk horn clarkia

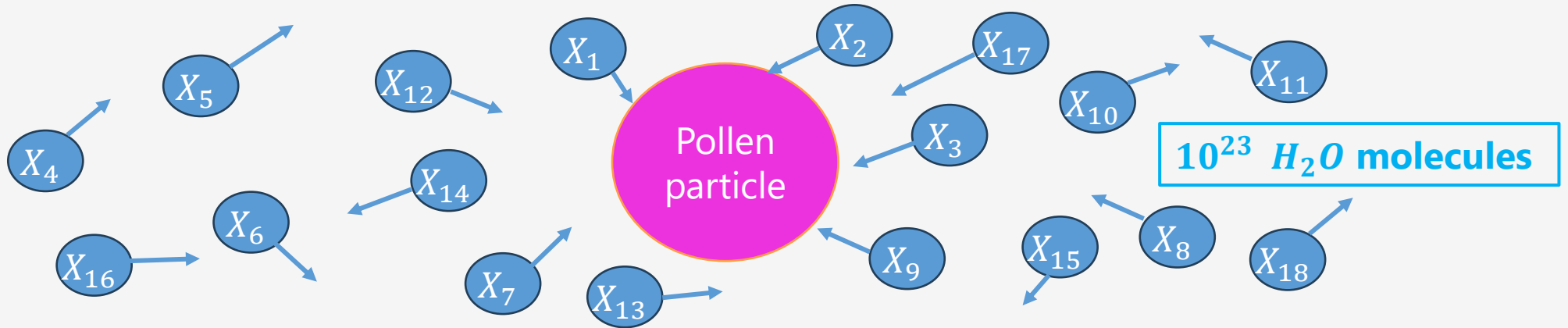


- There are **zillions molecules** in H_2O (or in any other liquid)
- The molecules move at very **high speeds**
- Their **speeds** and **directions** are **random**
- The pollen particle is suspended in the liquid
- It is **bombarded** by extremely large number of molecules at every time unit
- Every **collision** causes the pollen particle to move at a random direction
- The **aggregate effect** of all these collisions makes the pollen particle **move randomly**

Einstein's explanation of BM - 1905



Elk horn clarkia



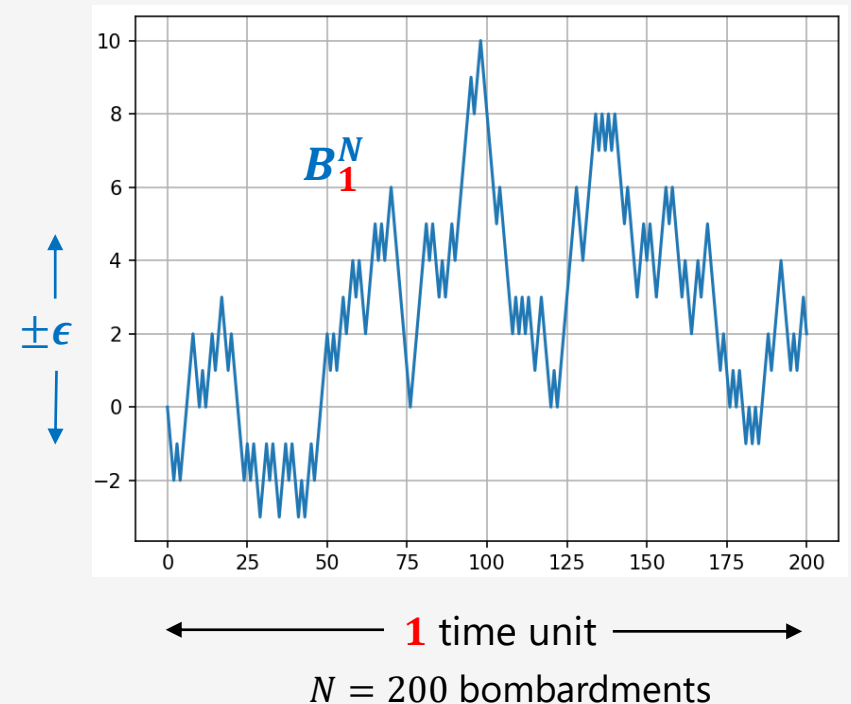
- S_0 = Initial position of the pollen particle
- X_k = displacement of pollen particle due to the bombardment of the **k -th molecule**
- $X_k \perp\!\!\!\perp X_\ell$ for all k, ℓ (no collusion among molecules!)
- Position of pollen particle after N bombardments
$$S_N = S_0 + X_1 + X_2 + X_3 + \cdots + X_N$$
- So, **Einstein's model** suggested that the pollen particle follows a **Random Walk**
- It also explained that **matter consists of molecules** (but this is more relevant for **physicists**)

BM – mathematical formulation

- Let's consider a **Simple & Symmetric RW** (for the pollen particle)
 - $P\{X_k = 1\} = P\{X_k = -1\} = 1/2$ and $X_k \perp\!\!\!\perp X_\ell \quad \forall k, \ell$ (**i.i.d.**)
 - $S_0 = 0 \rightarrow$ Starts at the **origin** (at $t = 0$), $S_0 \perp\!\!\!\perp X_k, \forall k$
 - There are N **bombardments per unit of time**
 - ϵ is the **size of the displacement** of the pollen particle
- The **position** of the pollen particle at time t is given by:

$$B_t^N = \epsilon X_1 + \epsilon X_2 + \cdots + \epsilon X_{tN}$$

Random variable;
Depends on N



BM as Continuous Time Limit (CTL) of a RW

$$B_t^N = \epsilon X_1 + \epsilon X_2 + \cdots + \epsilon X_{tN}$$

- To take the **CTL** we will need to take $N \rightarrow \infty$ and $\epsilon \rightarrow 0$ **(this can be tricky...)**
- We will **fix** $\epsilon > 0$ to a very small number that **depends on N** , and then **let** $N \rightarrow \infty$
 - Because the pollen particle should exhibit some small movement!
- We **fix** the **mean square displacement** *per time unit*

$$E[(\underbrace{B_1^N - B_0^N}_{\substack{\text{Displacement} \\ \text{in } 1 \text{ time unit}}})^2] = E[(B_1^N)^2] = \overset{\substack{\uparrow \\ \text{Fixed}}}{\sigma^2} > 0$$

By fixing σ^2 we ensure the particle always makes minute random movements

$$E[(B_1^N - B_0^N)^2] = E[(B_1^N)^2] = \sigma^2 > 0$$

- This implies that $\text{Var}(\mathbf{B}_1^N) = \sigma^2$, **because:** $\text{Var}(B_1^N) = E[(B_1^N)^2] - (E[B_1^N])^2$ **and** $E[\mathbf{B}_1^N] = 0$
- Let's show that $E[\mathbf{B}_1^N] = 0$

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- Let's show that $E[\mathbf{B}_1^N] = 0$
 - **Proof:** We know that all $X_1, X_2, \dots, X_N$ are **i.i.d.** and $P\{X_k = 1\} = P\{X_k = -1\} = 1/2$
 - Then we have: $E[X_k] = 1 \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0$ and $E[X_k^2] = 1^2 \cdot \frac{1}{2} + (-1)^2 \cdot \frac{1}{2} = 1$
 - From the definition of $\mathbf{B}_1^N$ we then have: $E[\mathbf{B}_1^N] = E[\sum_{k=1}^N \epsilon X_k] = \epsilon \sum_{k=1}^N E[X_k] = 0$
- **Remark:** We also have: $\text{Var}(X_k) = E[X_k^2] - (E[X_k])^2 = E[X_k^2] - 0 = 1$

- **Hence:** $\sigma^2 = \text{Var}(B_1^N) = \text{Var}\left(\sum_{k=1}^N \epsilon X_k\right) = \epsilon^2 \text{Var}\left(\sum_{k=1}^N X_k\right) = \epsilon^2 \sum_{k=1}^N \text{Var}(X_k) = \epsilon^2 N$

- This implies that **we must select the size of the displacement as** $\epsilon = \sigma/\sqrt{N}$

- Then the **position of the pollen particle** after t time units is As $N \rightarrow \infty$, the displacement $\epsilon \rightarrow 0$
at a very specific rate: $\sigma/\sqrt{N}$

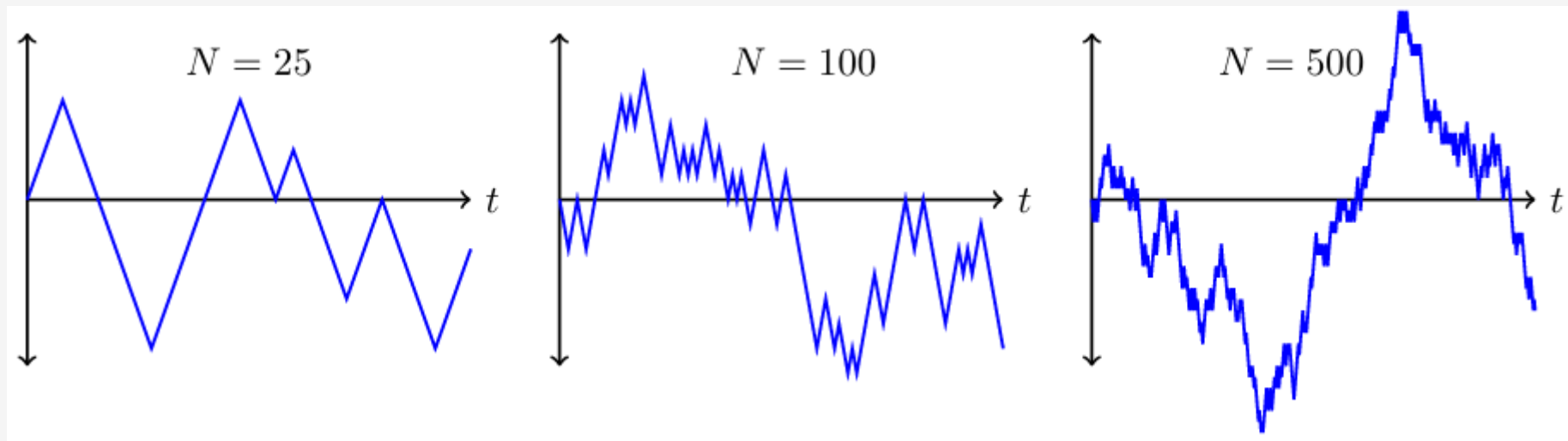
$$B_t^N = \frac{\sigma}{\sqrt{N}} (X_1 + X_2 + \cdots + X_{tN})$$

- **For different values of N , we get different Random Walks $\{B_t^N\}$**

- For larger values of N the paths of the RWs become closer to what Brown observed

$$B_t^N = \frac{\sigma}{\sqrt{N}} (X_1 + X_2 + \cdots + X_{tN})$$

- The **larger** N becomes, the paths of the RW $\{B_t^N\}$ become closer to what **Robert Brown** observed



The variance $\sigma^2 = \epsilon/\sqrt{N}$ remains constant as we increase N , because we decrease ϵ

Definition: Brownian Motion

- **Definition:** A **Brownian motion** is the **random process** $\{B_t\}_{t \geq 0}$ where each B_t is derived by the **Continuous Time Limit of** B_t^N , that is

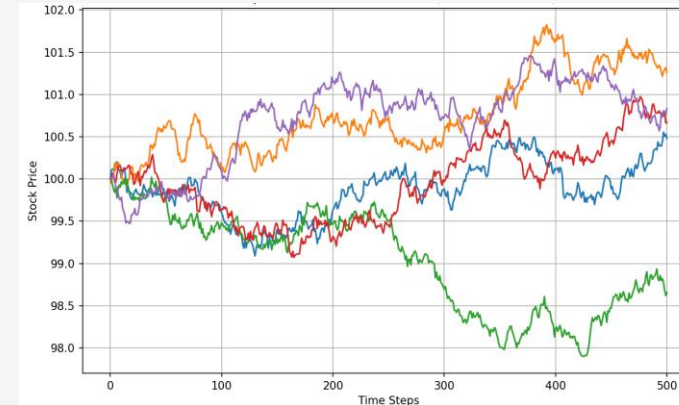
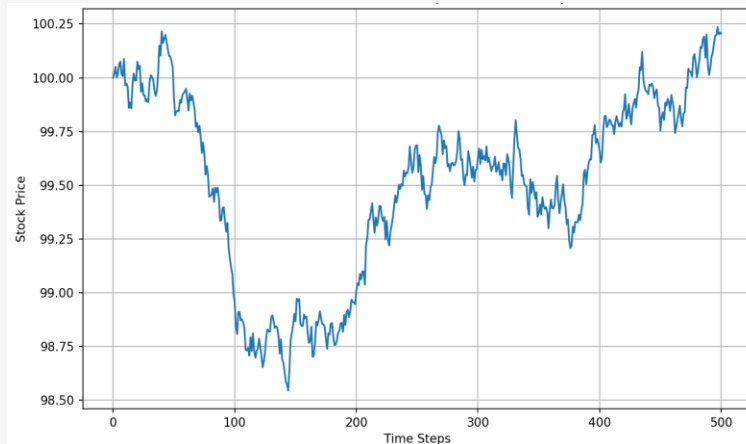
Random variable;
Does **not** depend on N

$$B_t = \lim_{N \rightarrow \infty} B_t^N = \lim_{N \rightarrow \infty} \frac{\sigma}{\sqrt{N}} (X_1 + X_2 + \dots + X_{tN})$$

- If $\sigma = 1$ then it is called standard BM
 - **The larger the σ is, the faster the particles move** (larger displacement)
- BM has **100s of applications** ranging from Biology and Physics, to Finance and Economics
 - Helps us describe how **molecules** find each other in **chemical reactions**
 - Helps us understand how **stock prices fluctuate** (and makes us gain or lose money...) 13

Modeling stock price movements

- In **financial modeling**, we often want to understand how stock prices change over time
- One approach is to think of **the stock price as evolving through a series of random steps**



The **stock price** is a result of **millions of small collisions** (i.e., trades, human decisions, news, etc.)

Multiplicative model

- **Example 1:** Assume you have a stock price where the price goes **up by $6/5^{th}$ s** with **probability $1/2$** and goes **down by $5/6^{th}$ s** with **probability $1/2$**
- Stock price at $n = 0$ is S_0 and the **stock price after n transactions:**

$$S_n = S_0 \prod_{i=1}^n Y_i$$

where Y_i are **i.i.d.** RVs with $P\{Y_i = 6/5\} = P\{Y_i = 5/6\} = 1/2, i = 1, 2, \dots, n$.

- But the **product** can be converted into a **sum** by taking **logarithms** on both sides...

Note that: $E[\log(Y_i)] = P\{Y_i = 6/5\} \log(6/5) + P\{Y_i = 5/6\} \log(5/6) = 0.5 \left(\log\left(\frac{6}{5}\right) + \log\left(\frac{5}{6}\right) \right) = 0$

Convention: log is ln

$$S_n = S_0 \prod_{i=1}^n Y_i \Leftrightarrow \log S_n = \log S_0 + \log Y_1 + \cdots + \log Y_n$$

Sums are easier to work with than products

Note: we always denote by $\log(x)$ the **natural logarithm** of $x > 0$, that is $\log(x) \equiv \ln(x)$

where each random term is defined by the RV:

$$\log Y_i = \begin{cases} \log 6/5, & \text{with } p = 1/2 \\ -\log 6/5, & \text{with } p = 1/2 \end{cases} \quad (\log 5/6 = -\log 6/5)$$

- Define the RV: $X_i \in \{1, -1\}$ with $P\{X_i = 1\} = P\{X_i = -1\} = 1/2$, then $\log Y_i = (\log 6/5)X_i$

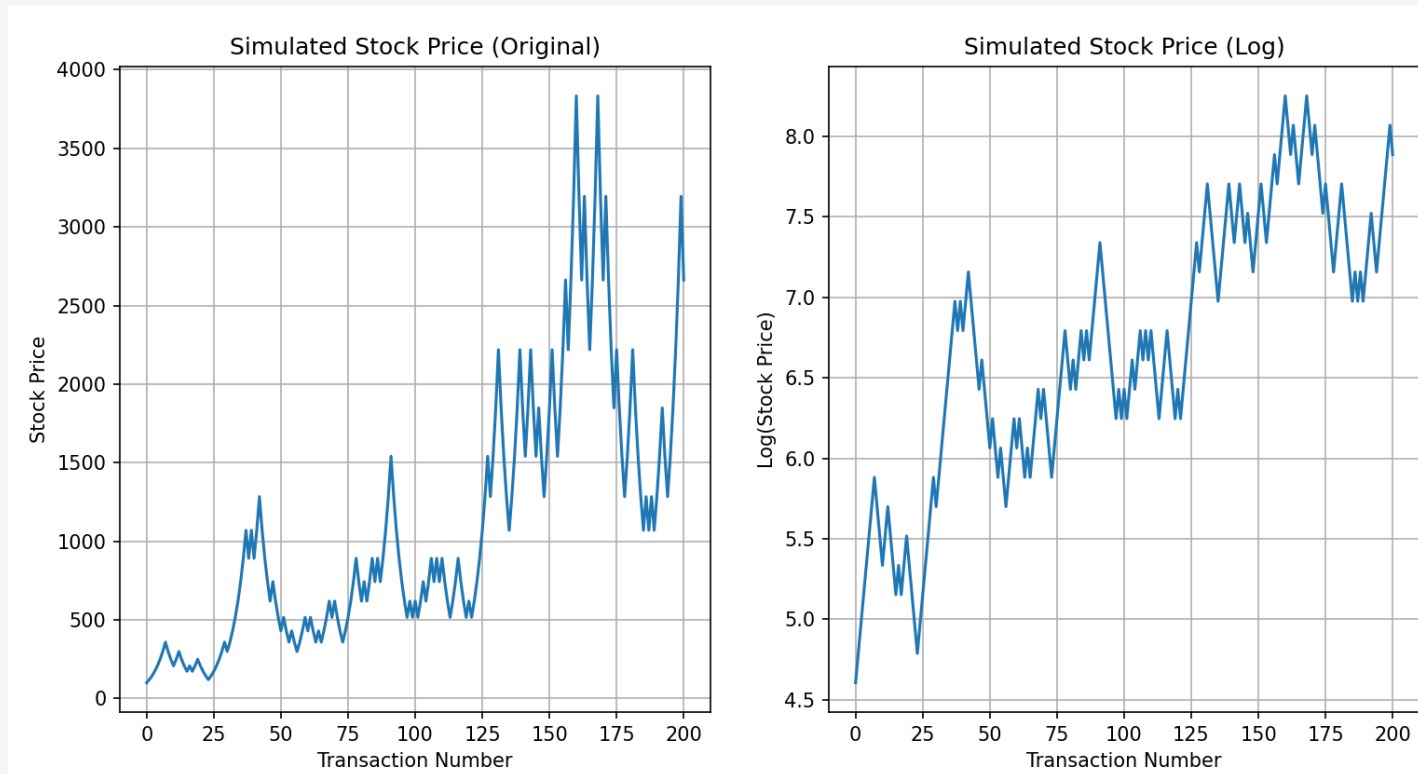
$$\log S_n = \log S_0 + (\log 6/5)X_1 + \cdots + (\log 6/5)X_n \Rightarrow$$

$$\Rightarrow \log(S_n/S_0) = (\log 6/5)X_1 + \cdots + (\log 6/5)X_n$$

It is a Random Walk

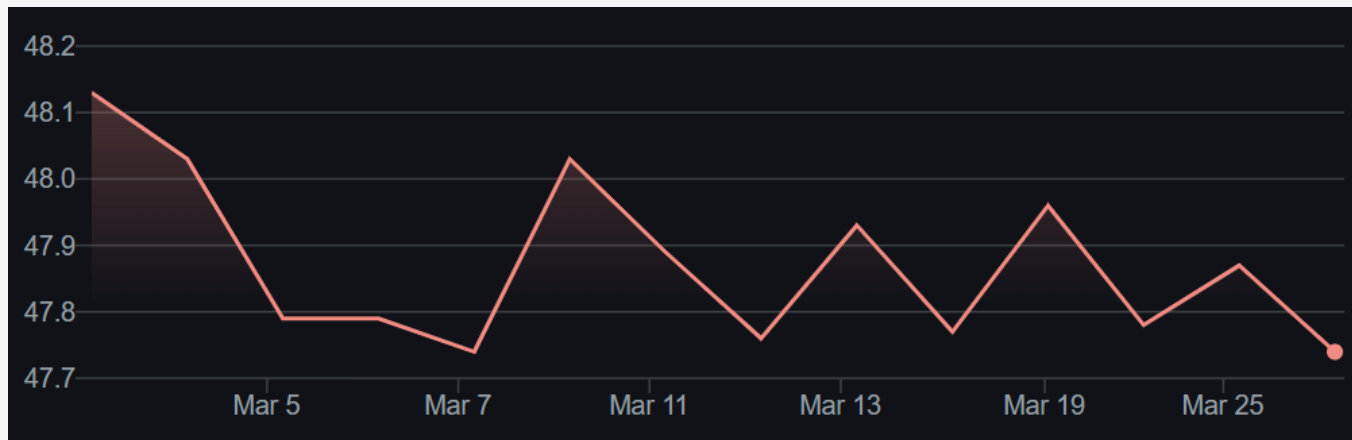
**Cumulative log return
(from 0 to n)**

S_n vs $\log S_n$



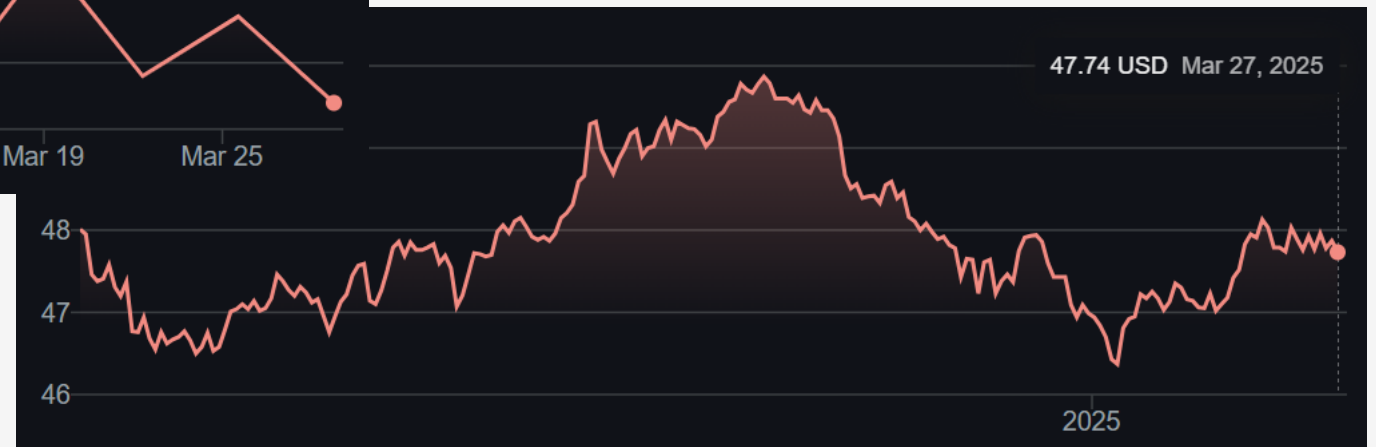
For stocks or ETFs with **very low transaction volume** we can see the RW of the price

For example, the **US Treasury 7 Year Note ETF** has **838 pieces** average volume



USVN

1 week



1 year

Another multiplicative model

- **Example 2:** Assume you have a stock price where the price goes **up** by $6/5^{ths}$ with probability $1/2$ and goes **down** by $5/6^{ths}$ with probability $1/2$
- Stock price at $n = 0$ is S_0 and the stock price after n transactions:

$$S_n = S_0 \prod_{i=1}^n \left(\frac{6}{5}\right)^{X_i} \left(\frac{5}{6}\right)^{1-X_i}$$

where X_i are **independent Bernoulli RVs** with $P\{X_i = 1\} = p$ and $P\{X_i = -1\} = 1 - p$

Simplifying the model with logarithms

- To make the Multiplicative model easier to analyze, we take the logarithm of the stock price

$$\log(S_n) = \log(S_0) + \sum_{i=1}^N \left(X_i \log\left(\frac{6}{5}\right) + (1 - X_i) \log\left(\frac{5}{6}\right) \right)$$

- The logarithm of a stock price can be modeled as a simple symmetric RW.
- Each step in the RW is $\log(Y_i)$, representing a random increment